WEEK 2 · DAY 2, PART 2 · OVERVIEW

pandas

Series and DataFrames, Part 2



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ABOUT 55 MIN WITH EXERCISES / 5 SECTIONS / 7 EXERCISES

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The Desk Lead's Eleventh Note

The file is loaded; now make it answer. What is in the book, sector by sector? Which bonds pass a test: Energy, below par, long and cheap? The widest spreads, the biggest positions. Mark BEZL 99039NAC0 at 97.00 on a copy, not our record, and tell me what it costs. Then send the widest spreads in a table I can forward.

ABOUT 55 MIN WITH EXERCISES / 5 SECTIONS / 7 EXERCISES

Your Route Through Day 2, Part 2

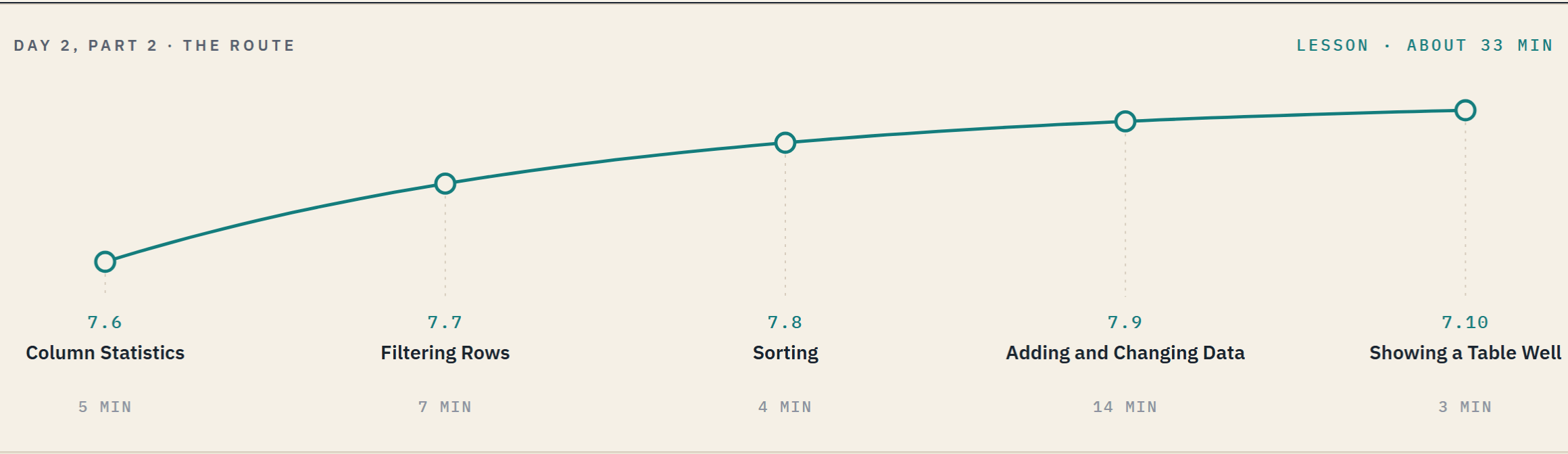
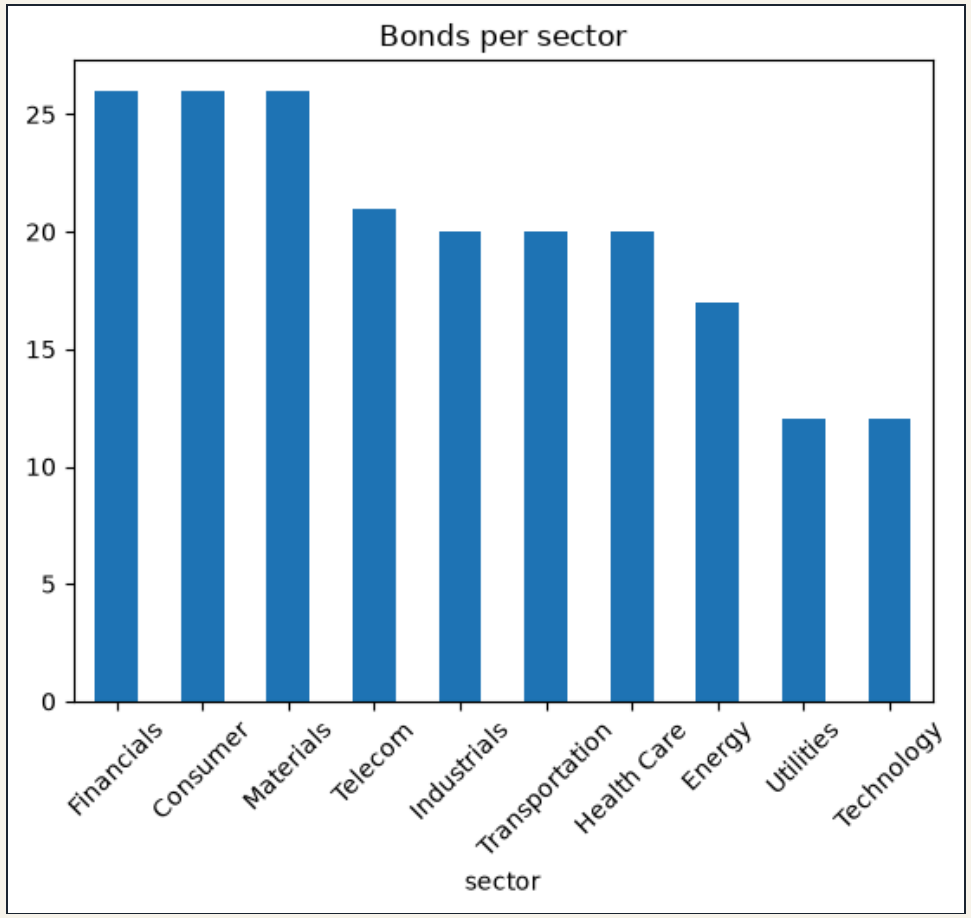


FIGURE 1 Five sections, in order, with about how long each takes.

Time. About 33 minutes for the lesson, then about 21 for the seven exercises.

7.6 Column Statistics



TOTALS AND THE MIDDLE

```
print('Market value:', holdings['market_value'].sum())
print('Face held:', holdings['par_held'].sum())
print('OAS mean:', holdings['oas'].mean(), 'median:', holdings['oas'].median())
print('Price low:', holdings['price'].min(), 'high:', holdings['price'].max())
```

```
Market value: 490819215.0
Face held: 489000000
OAS mean: 185.75 median: 135.5
Price low: 62.045 high: 119.131
```

EXCEL SIDE BY SIDE

PANDAS	EXCEL
`holdings['market_value'].sum()`	`=SUM(S2:S201)`
`holdings['oas'].mean()`	`=AVERAGE(M2:M201)`

By count of bonds, not by money: market value by sector is the next notebook's question.

7.7 Filtering Rows

holdings[is_energy]: the True rows, each keeping its label

label	ticker	sector	== 'Energy'
65	FESR	Technology	False
66	FESR	Technology	False
67	GALN	Energy	True
68	GALN	Energy	True

rows 69 to 72: four more False

73	GALY	Energy	True

holdings[is_energy]

67	GALN	Energy
68	GALN	Energy
73	GALY	Energy

67, 68, 73: no renumbering

FIGURE 2 A True or False test for Energy beside each row; the True rows come out as a table that keeps their labels.

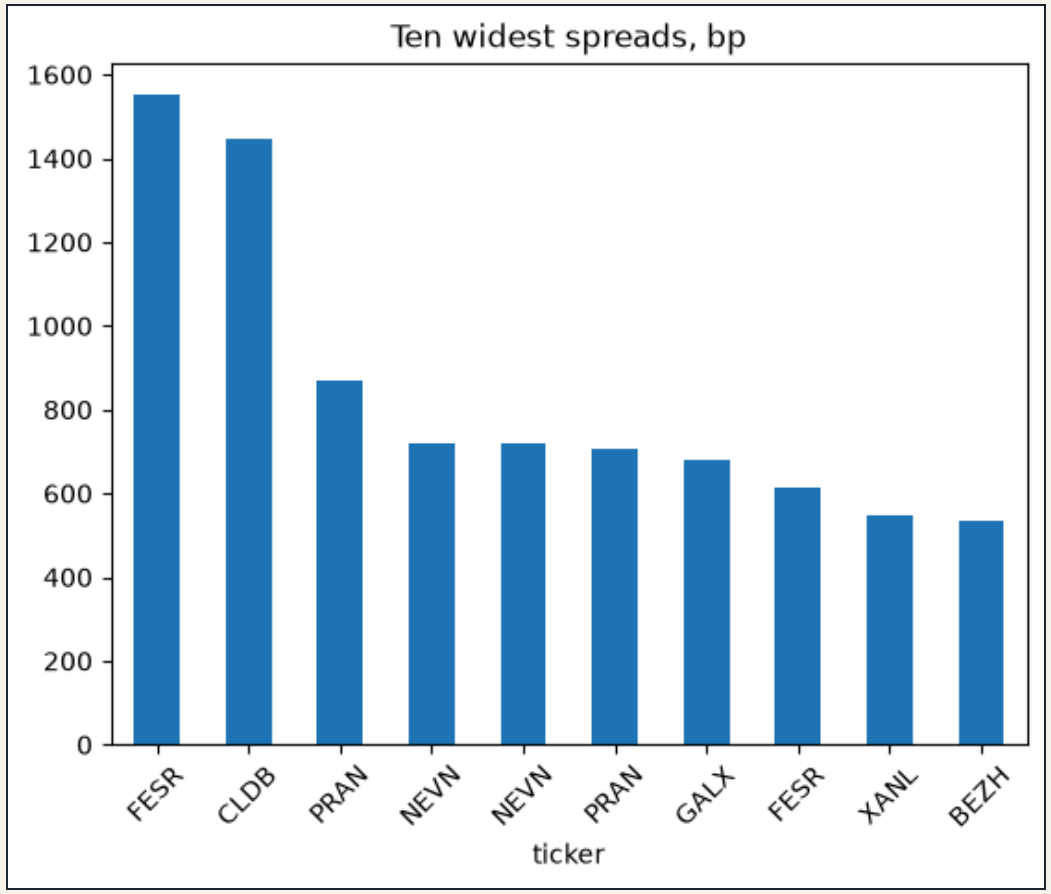
KEEP THE TRUE ROWS

```
energy = holdings[is_energy]
```

	TICKER	ISSUER
67	GALN	Galvostrin Resources
68	GALN	Galvostrin Resources
73	GALY	Galvolvey Resources
82	HESR	Heskentor Resources

A filter keeps each row's label.

7.8 Sorting



SORT, WIDEST FIRST

```
widest = holdings.sort_values('oas', ascending=False)
```

	TICKER	SECTOR	RATING	PRICE	OAS
65	FESR	Technology	CCC+	87.116	1551
17	CLDB	Consumer	B	62.045	1446
132	PRAN	Financials	CCC+	89.407	871
110	NEVN	Utilities	B-	96.616	720
111	NEVN	Utilities	B-	94.053	719

EXCEL SIDE BY SIDE

Data, Sort reorders the sheet itself; `=SORT(A2:S201, 13, -1)` in Excel 2021 and Microsoft 365 spills a sorted copy, as `sort_values()` hands one back.

FESR, PRAN, and NEVN each appear twice: a ticker is not a bond.

7.9 Adding and Changing Data

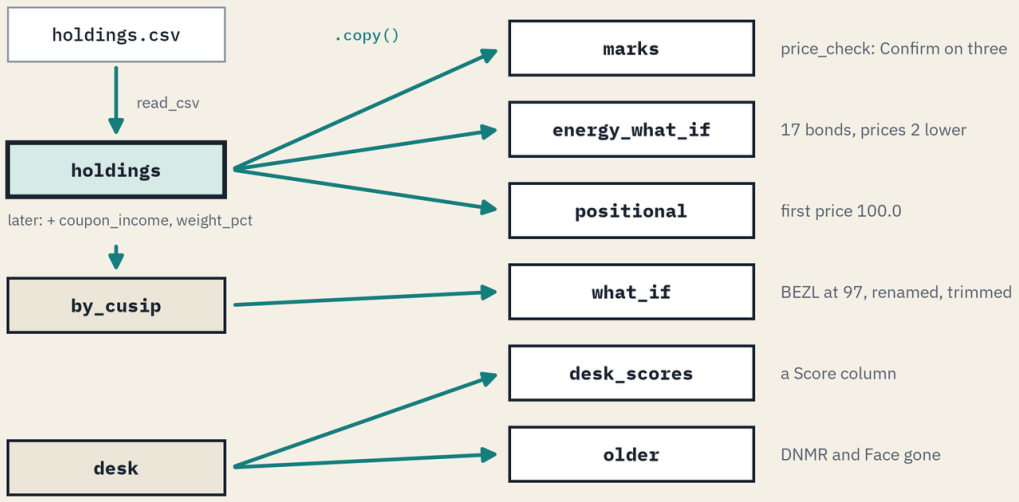


FIGURE 6 Six tables made from copies, each with its own change; holdings.csv and holdings keep every value.

CHANGE A COPY

```
what_if = by_cusip.copy()
what_if.loc['99039NAC0', 'price'] = 97.0
print('What-if:', what_if.loc['99039NAC0', 'price'])
print('Record:', by_cusip.loc['99039NAC0', 'price'])
```

```
What-if: 97.0
Record: 98.73
```

MARKED FOR THE PRICING TEAM

	TICKER	PRICE	PRICE_CHECK
17	CLDB	62.045	Confirm
117	NEVX	73.873	Confirm
150	SYLX	76.903	Confirm

Every what-if landed on a copy. Now show it well.

Market Value Does Not Follow

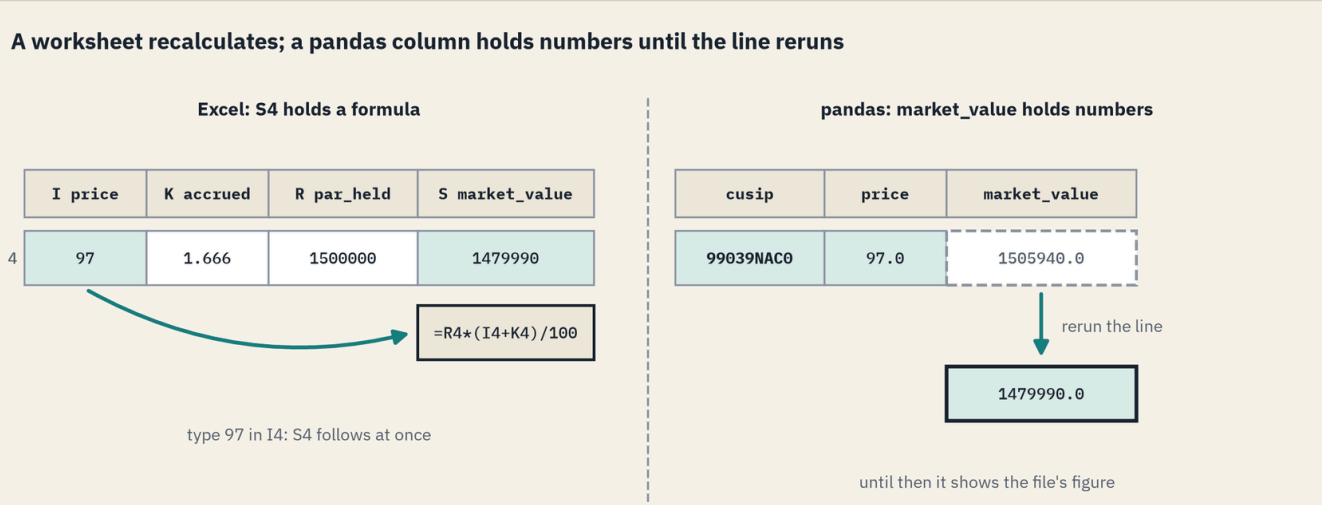


FIGURE 5 Excel's formula updates the moment 97 is typed; the pandas market value waits until the line reruns.

STILL THE FILE'S FIGURE

```
print(what_if.loc['99039NAC0', 'market_value'])
```

```
1505940.0
```

AFTER THE RERUN

```
print('Change in market value:', round(change, 2))
```

```
Change in market value: -25950.0
```

A column holds values, not formulas: change an input, rerun the line.

7.10 Showing a Table Well

The five widest spreads: as printed, and ready to send

as printed

	ticker	rating	price	oas	market_value
65	FESR	CCC+	87.116	1551	2216725.0
17	CLDB	B	62.045	1446	1951590.0
132	PRAN	CCC+	89.407	871	3889685.0
110	NEVN	B-	96.616	720	1214187.5
111	NEVN	B-	94.053	719	4121437.5

row labels, raw decimals, no unit, no title

styled

Five widest spreads

ticker	rating	price	oas	market_value
FESR	CCC+	87.12	1,551 bp	2,216,725
CLDB	B	62.05	1,446 bp	1,951,590
PRAN	CCC+	89.41	871 bp	3,889,685
NEVN	B-	96.62	720 bp	1,214,188
NEVN	B-	94.05	719 bp	4,121,438

every number as the desk writes it

FIGURE 7 The five widest spreads as pandas prints them, then styled with a caption, formats, shading, and bars.

SHADE, MARK, AND BAR

```
    .background_gradient(subset=['oas'], cmap='Greens')
    .highlight_max(subset=['price'], color='#f4d58d')
    .bar(subset=['market_value'], color='#8fc9ae')
    .set_caption('Five widest spreads')
    .hide(axis='index'))
```

EXCEL SIDE BY SIDE

PANDAS	EXCEL
`.background_gradient(cmap='Greens')`	Conditional Formatting, Color Scales

For another reader: formatted numbers, a caption, shading only where it helps.

You Can Now

- Total, average, and count a column, with the median beside the mean.
- Count each group and its share with **`value_counts()`**.
- Filter the book on one test or several, and count or total what passes.
- Rank it: widest spreads, largest positions.
- Add a calculated column, and run a what-if on a copy.
- Set every row that passes a test with **`loc[mask, column] = value`**, copying a filtered table first.
- Drop columns and rows, renumber the index, and show a table with formats, shading, and bars.

NEXT
WEEK 2 · DAY 3, PART 1

pandas: Combining and Loading, Part 1

Joining the desk's tables together, stacking them, and summarizing by group.